Spatial and land use considerations of high renewable penetration into energy systems

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Supervisor: Dr. Taco Niet
Aug 4th, 2026

Outline

Land Acknowledgement

Introduction

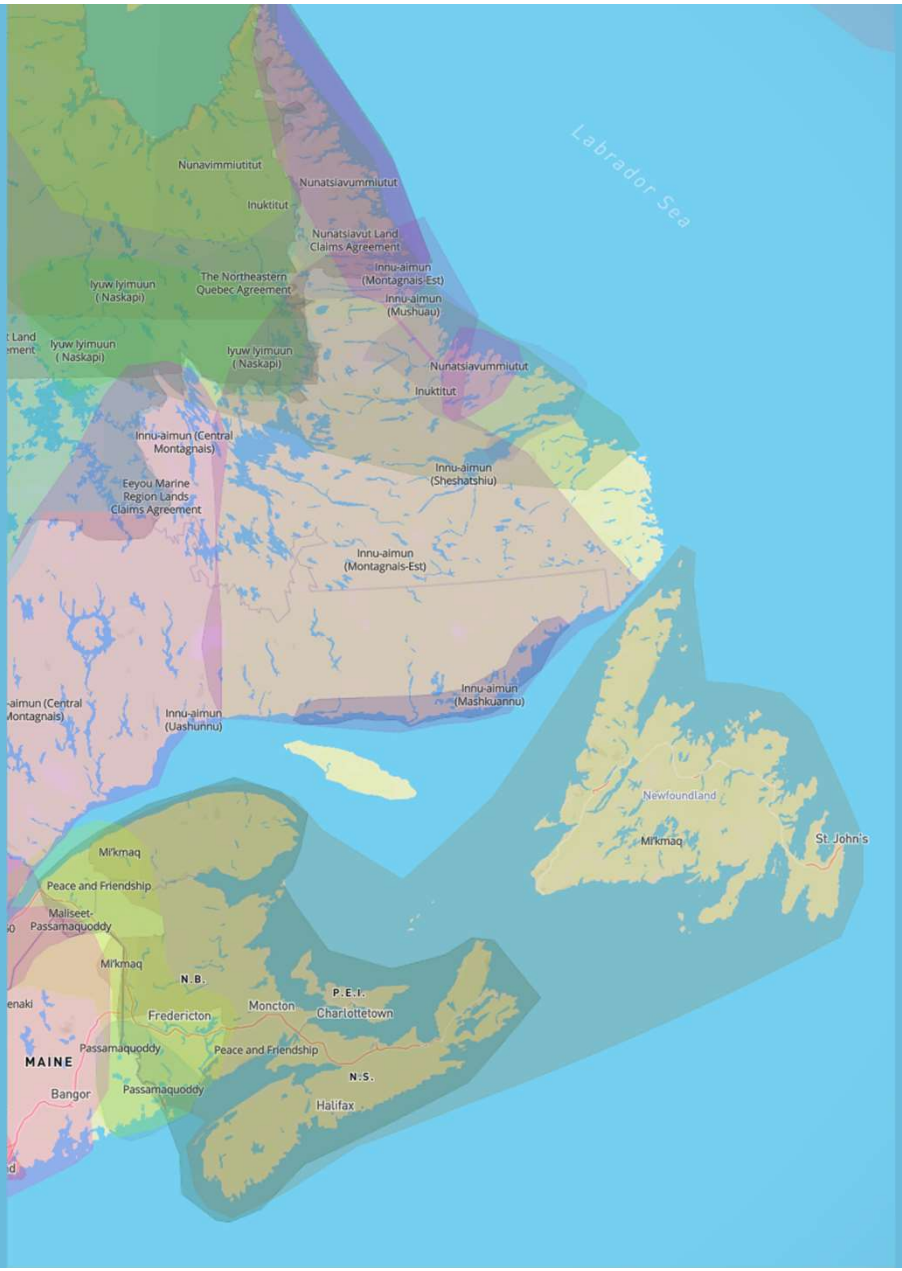
Methods and Tools

Case Study

- **Canadian provinces**
- **Atlantic provinces**
- **Prince Edward Island**

Conclusions

Land Acknowledgement



Source: native-land.ca





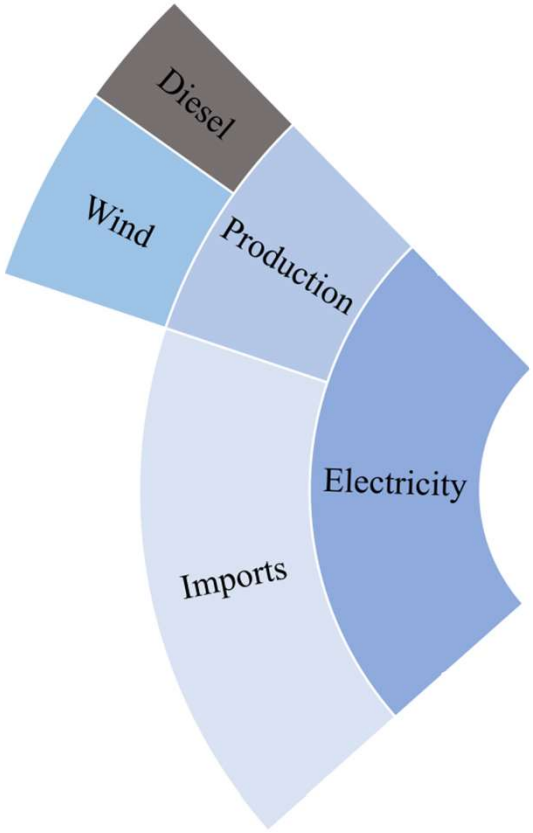
Introduction:

MOTIVATION

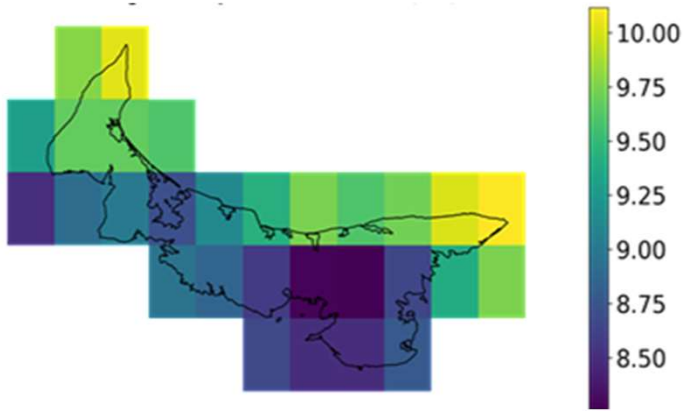
Net Zero Emissions

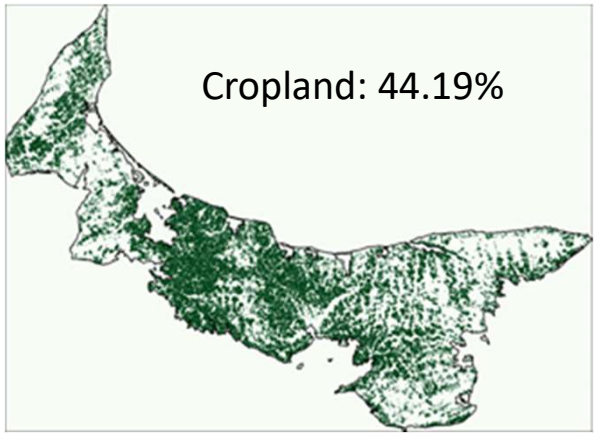
Renewable Penetration

Land Intensity



Average Wind Speed at 100 Meters (m/s)

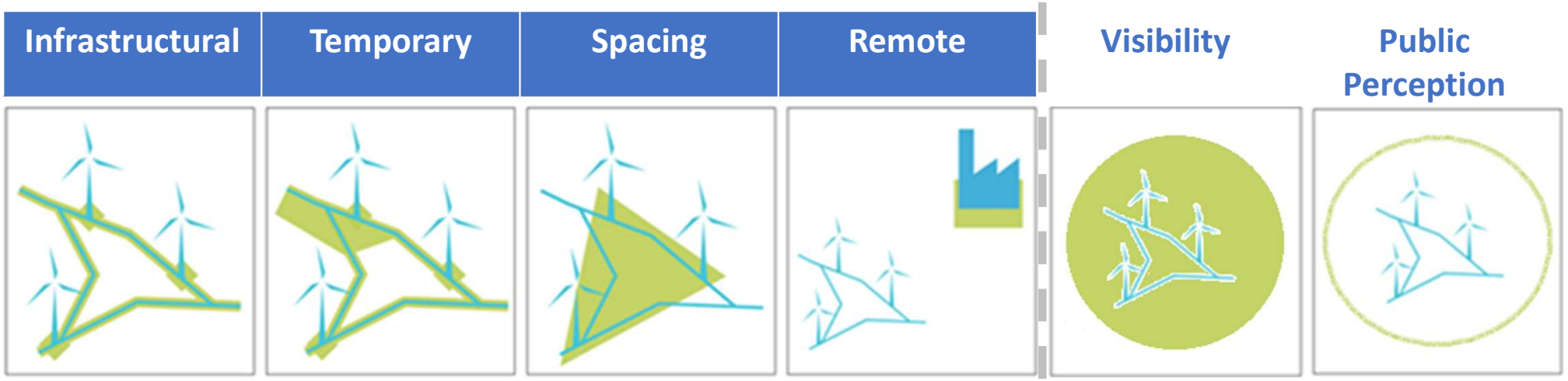




Note: Reference numbers are as they appear in the thesis

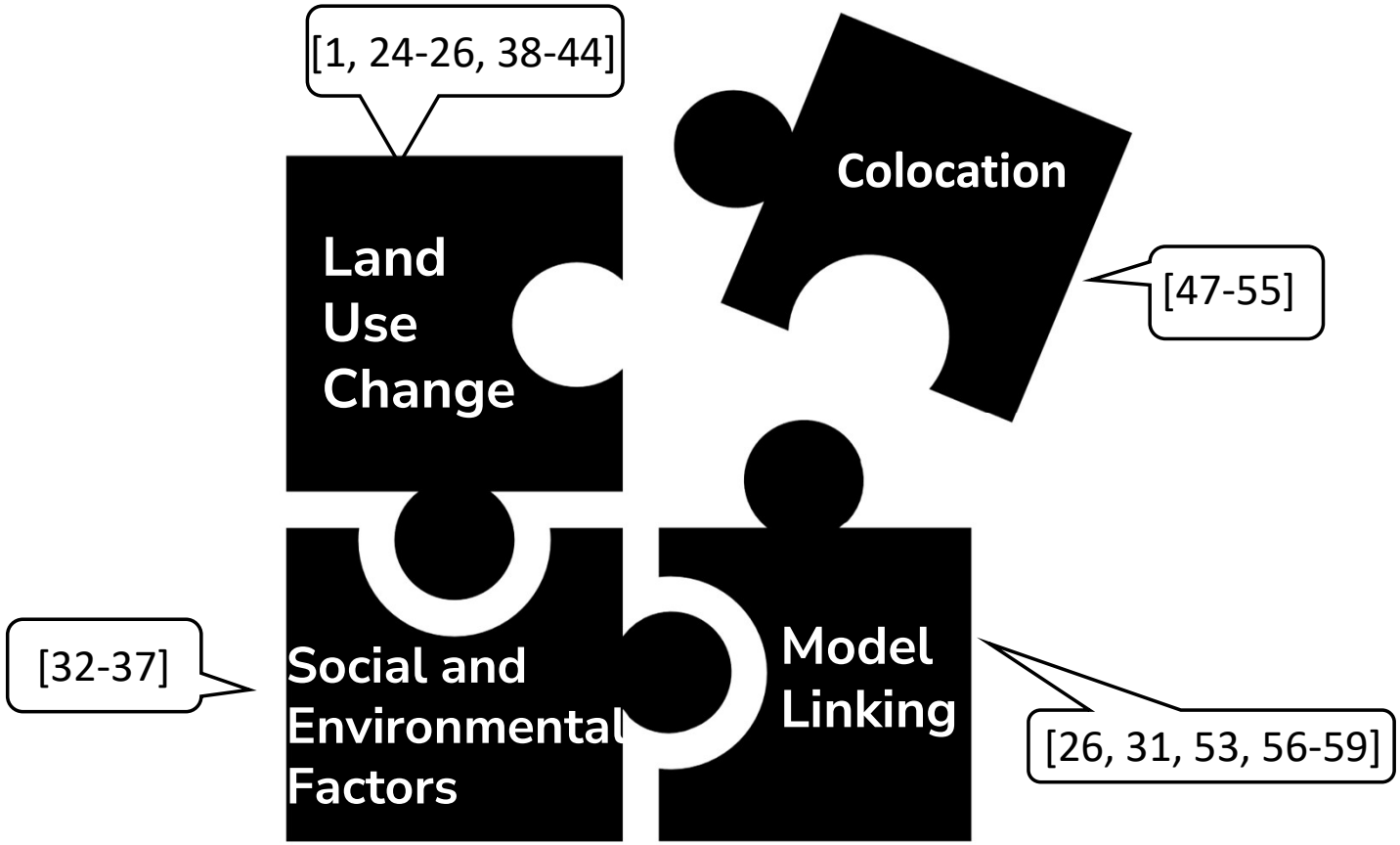
Source: [6, 9, 11]

Background



Source: [4]

Literature Gaps



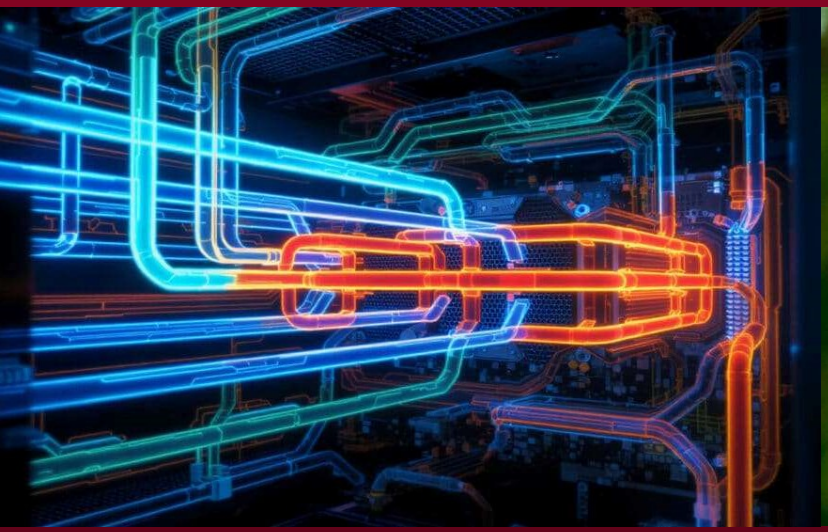
Research Questions

What Canadian land-cover layers overlap with onshore wind opportunities at the provincial scale?

How do GIS-derived, spatially explicit wind technologies affect model outputs?

How do regional policies impact the land prospect?

How do colocation modeling practices change the energy system's decision-making?

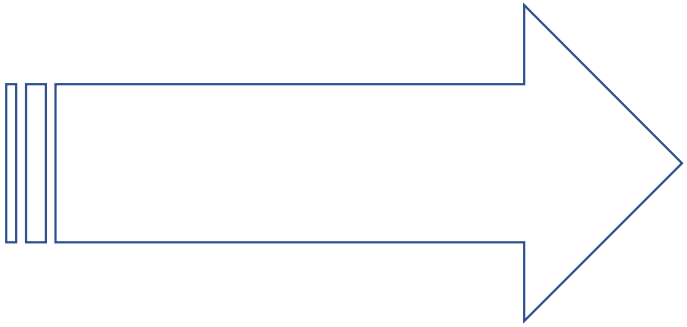




Methods:

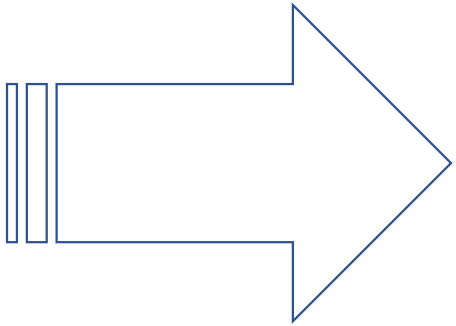
Model Development

1. Spatial Modeling
   - What type of land cover?
   - How much wind potential?
   - How far from the grid?



GIS+ Python(Atlite, etc.)

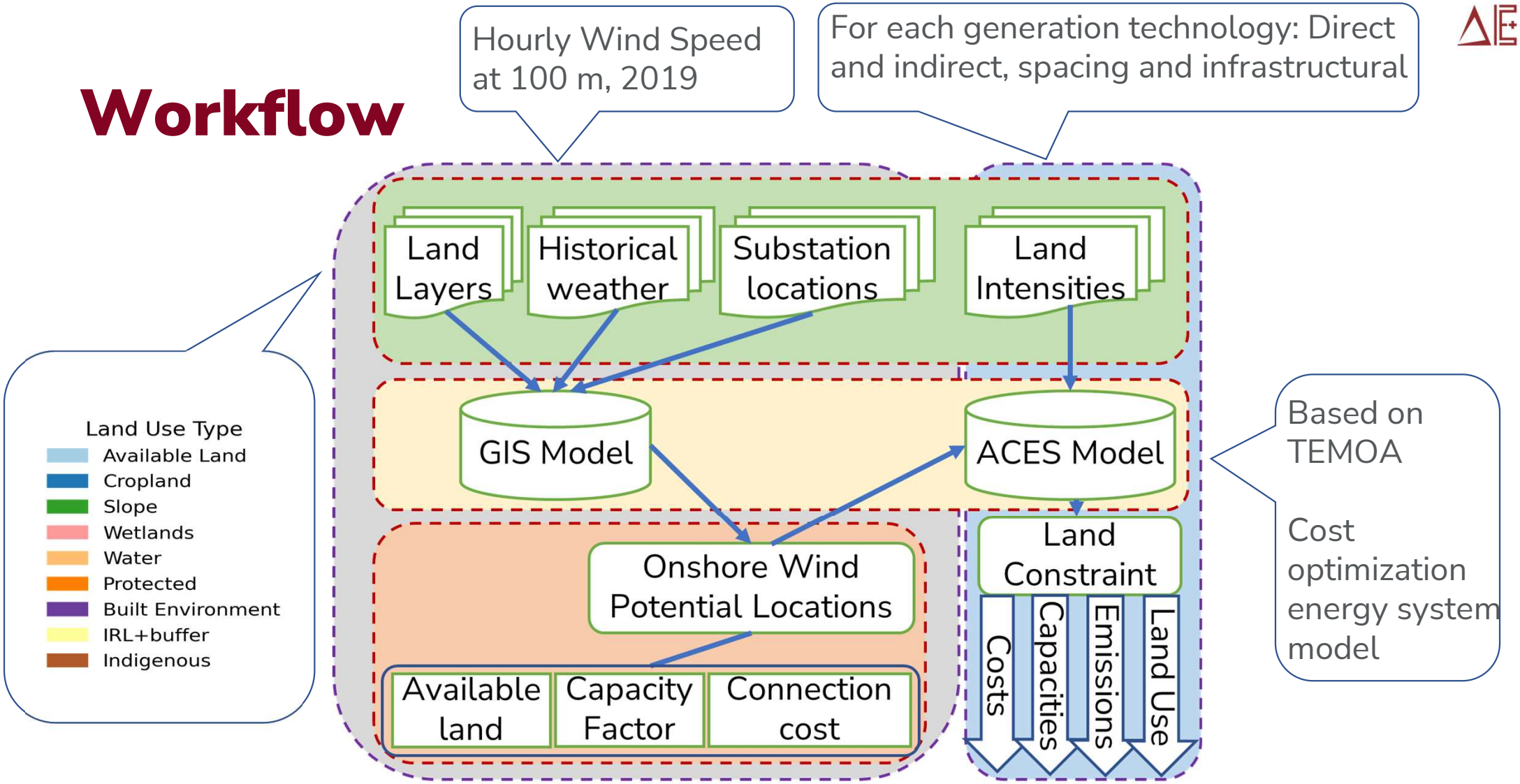
2. Land Tracking
   - What type of land use?
   - How much land use per technology?



TEMOA-ACES

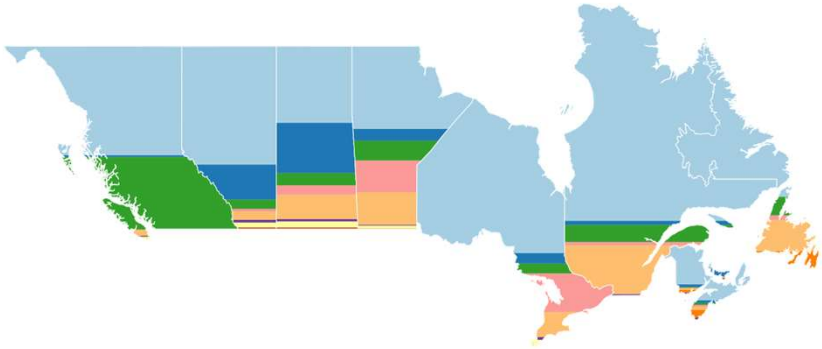
3. Land Limit
   - What are the trade-offs and synergies?

Workflow

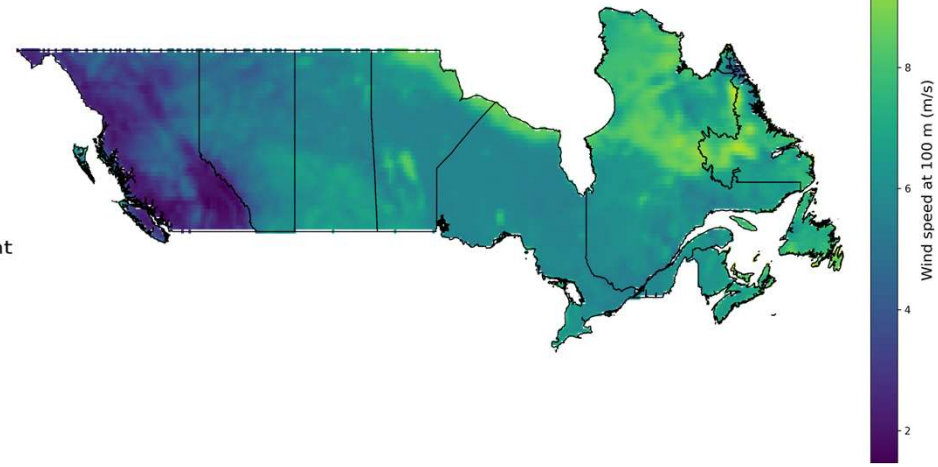

Hourly Wind Speed at 100 m, 2019
For each generation technology: Direct and indirect, spacing and infrastructural
Land Use Type
Available Land
Cropland
Slope
Wetlands
Water
Protected
Built Environment
IRL+buffer
Indigenous
Land Layers
Historical weather
Substation locations
Land Intensities
GIS Model
ACES Model
Onshore Wind Potential Locations
Available land
Capacity Factor
Connection cost
Land Constraint
Costs
Capacities
Emissions
Land Use
Based on TEMOA
Cost optimization energy system model

Scenario Overview

1. Land vs. Wind Constraints & Potentials



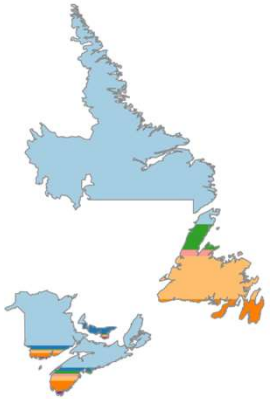




Scenario Overview

1. Land vs. Wind Constraints & Potentials
2. Spatial development impacts

TEMOA-ACES
Base(Spatially Enhanced TEMOA)



Scenario Overview

1. Land vs. Wind Constraints & Potentials
2. Spatial development impacts
3. Policy Impacts

Base(Spatially Enhanced TEMOA)	ITC
RPS	Mix



ITC: Investment Tax Credit
RPS: Renewable Portfolio Standard

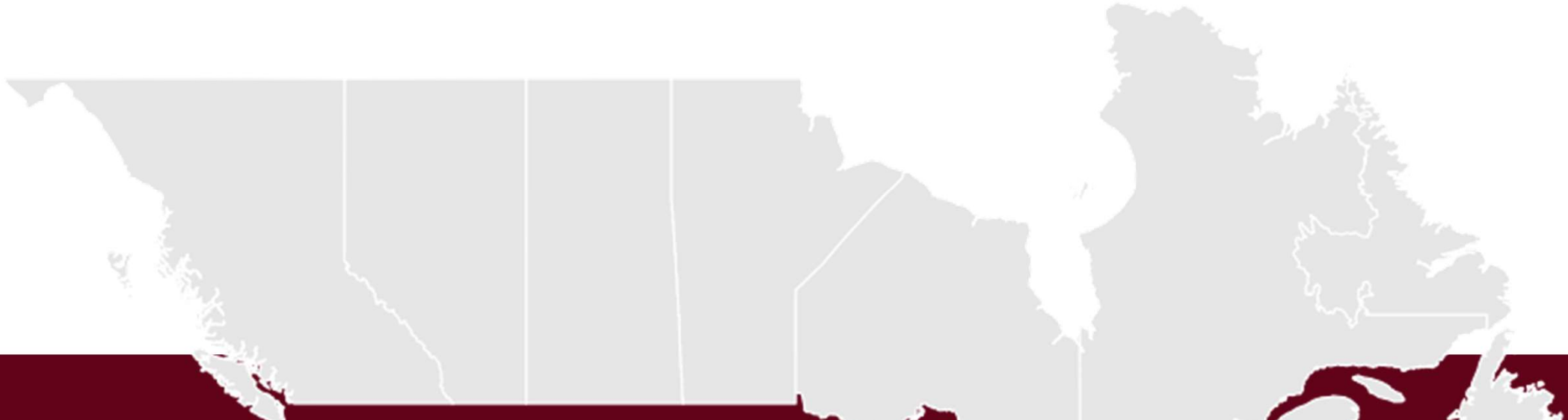
Scenario Overview

1. Land vs. Wind Constraints & Potentials
2. Spatial development impacts
3. Policy Impacts
4. Onshore Wind and Agriculture Colocation



Base(Spatially Enhanced TEMOA) Conventional					
Spacing					
Colocation					
Land Limits (km2)	-	100	50	5	1

SFU

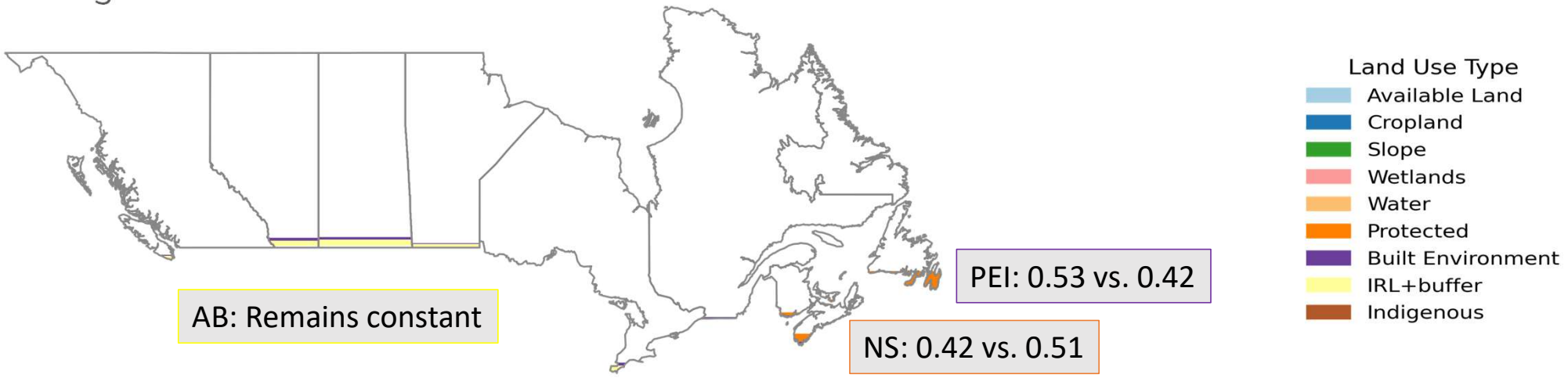


Case Study

Canada's Onshore Wind Deployment under Competing Land Uses

1. Low percentage land covers

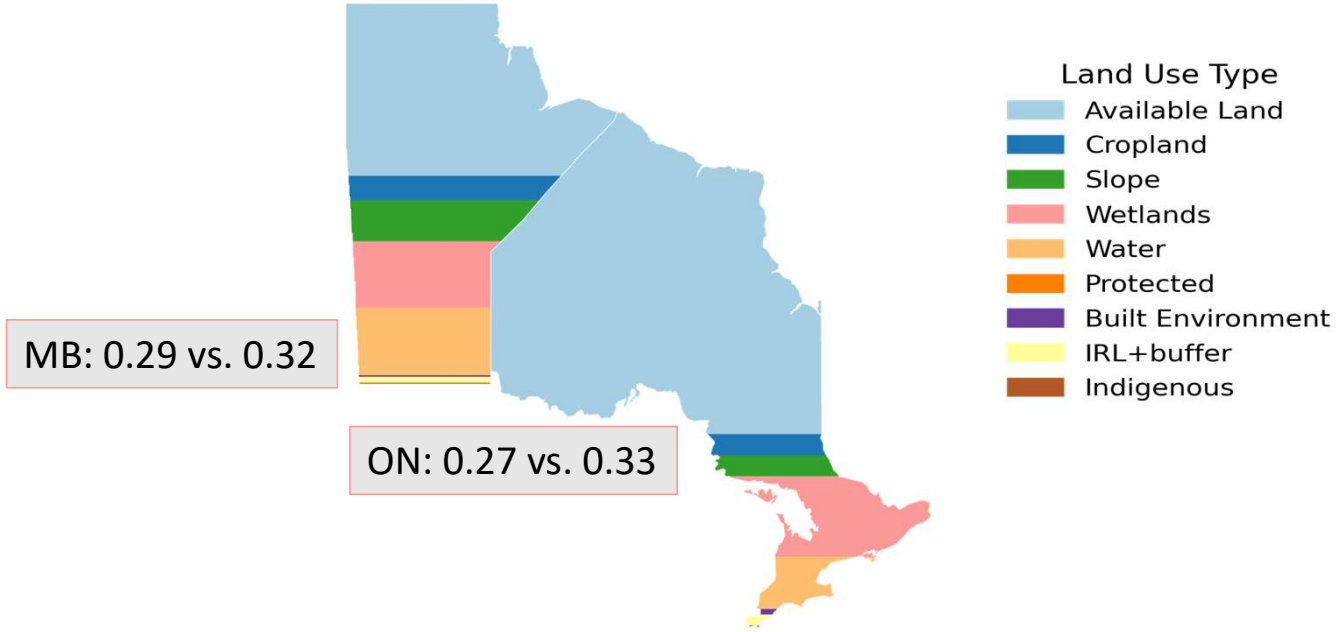
- Protected and Conserved Areas
- Built Environment
- Indigenous Reserve Lands +Buffer



Canada's Onshore Wind Deployment under Competing Land Uses

2. High land coverage, Low Wind Potential

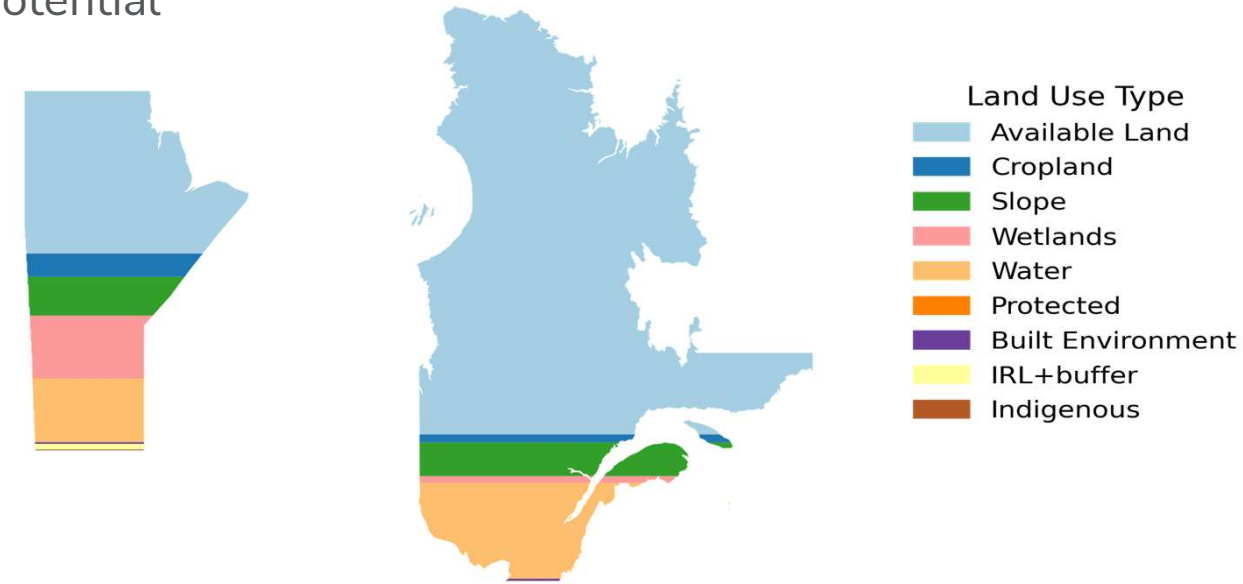
- Wetland



Canada's Onshore Wind Deployment under Competing Land Uses

2. High land coverage, Low Wind Potential

- Waterbodies

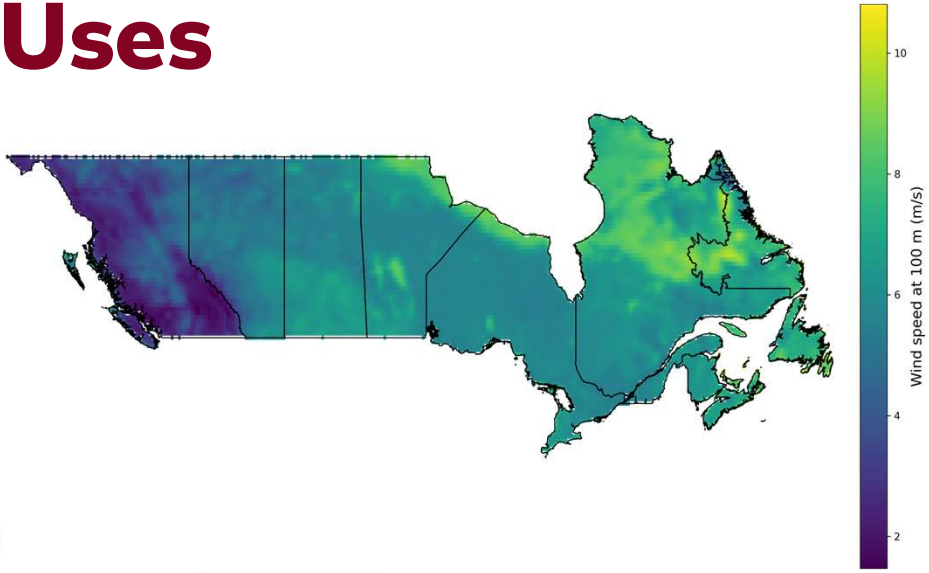


MB: 0.29 vs. 0.32

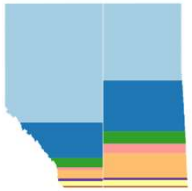
QC: 0.36 vs. 0.54

Canada's Onshore Wind Deployment under Competing Land Uses

3. High Land Coverage, High Wind Potentials
- Cropland







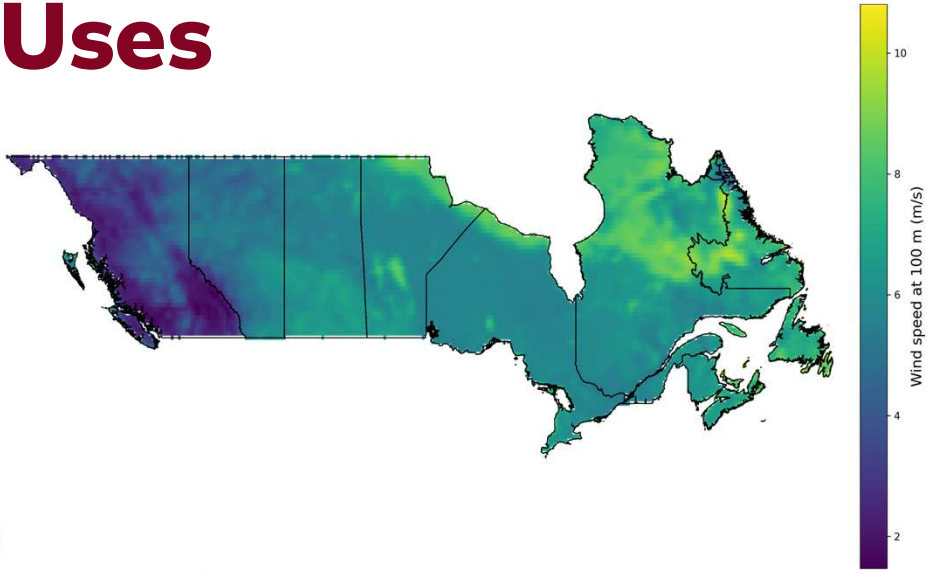
SK: 0.31 vs. 0.27

AB: 0.22 vs. 0.166

Canada's Onshore Wind Deployment under Competing Land Uses

3. High Land Coverage, High Wind Potentials

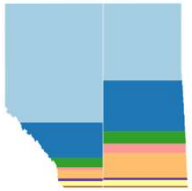
- Cropland- PEI





PEI: 0.49 vs. 0.52



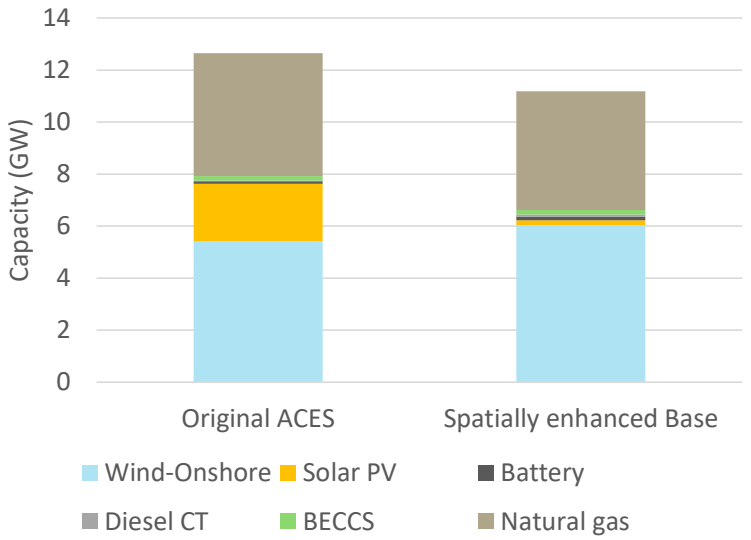


SK: 0.31 vs. 0.27

AB: 0.22 vs. 0.166

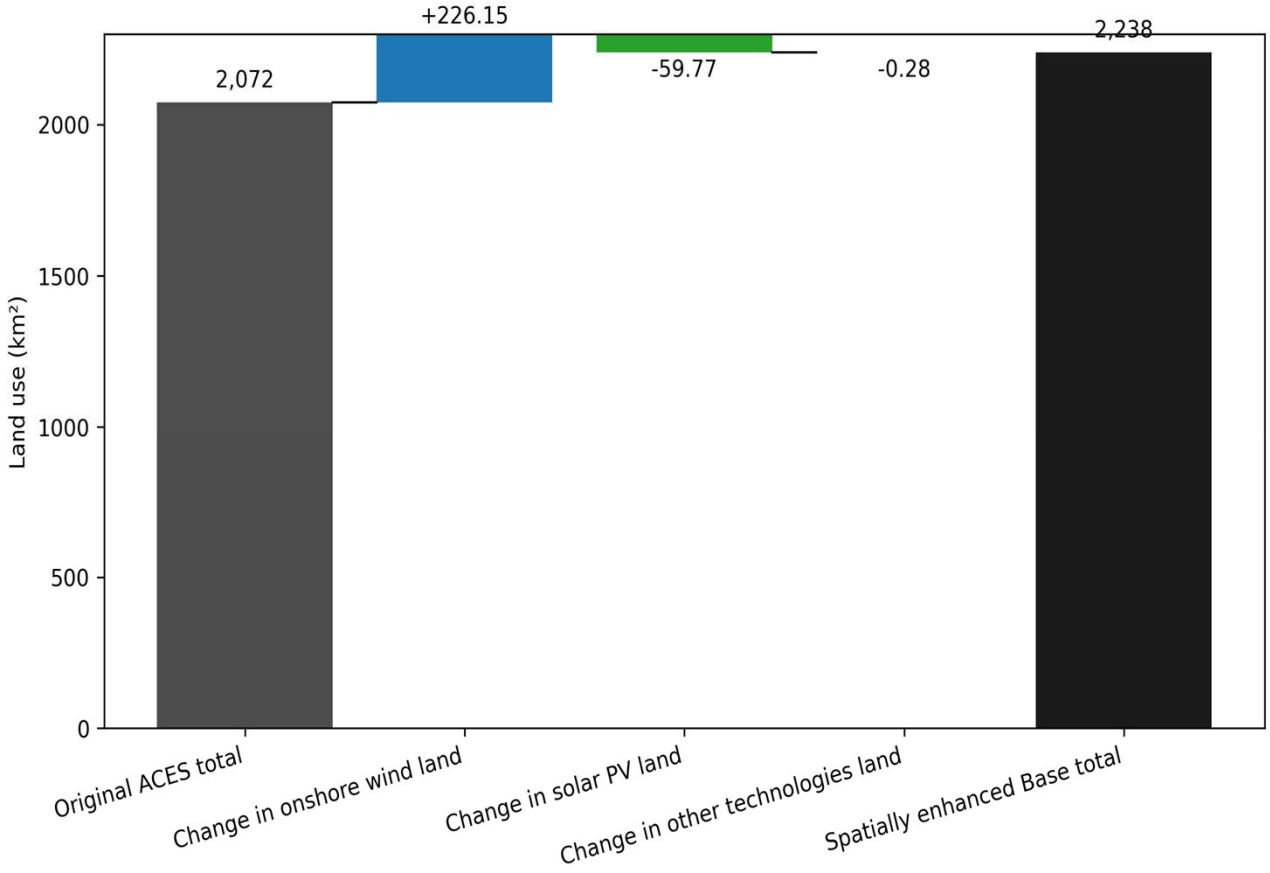
Spatially-enhanced TEMOA-ACES model

- Higher Capacity Factors
- Lower Total Capacity
- More onshore wind
- Lower solar PV buildout



Spatially-enhanced TEMOA-ACES model

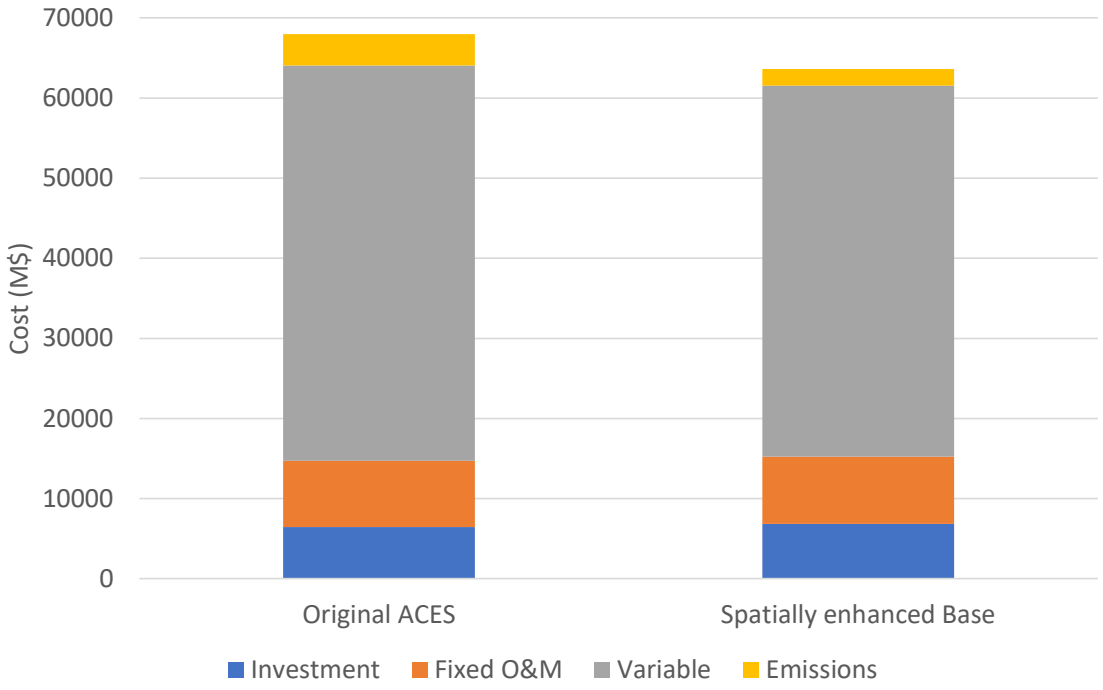
- More Onshore Wind
- More Land Use



Spatially-enhanced TEMOA-ACES model

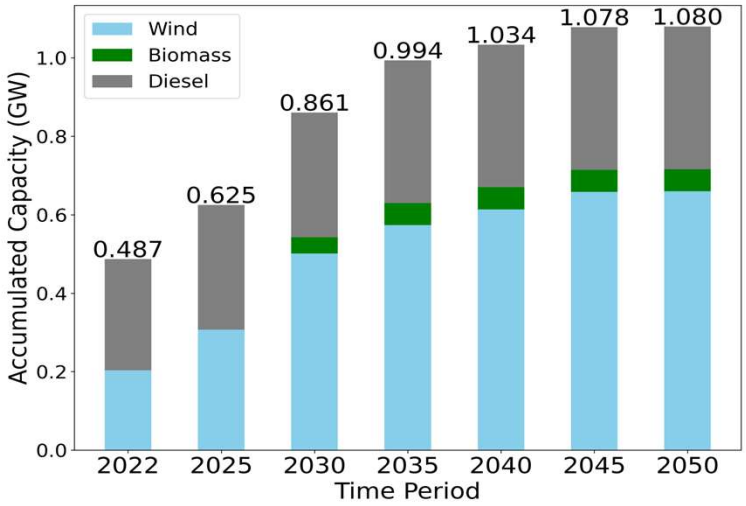
- Higher Transmission Costs
- Lower Overall Cost

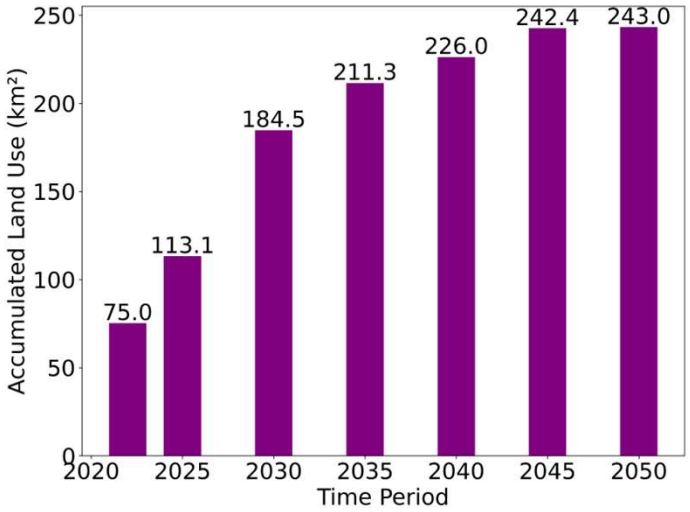
Due to higher capacity factors



How different policies impact PEI's future Energy and Land Use- Base Scenario

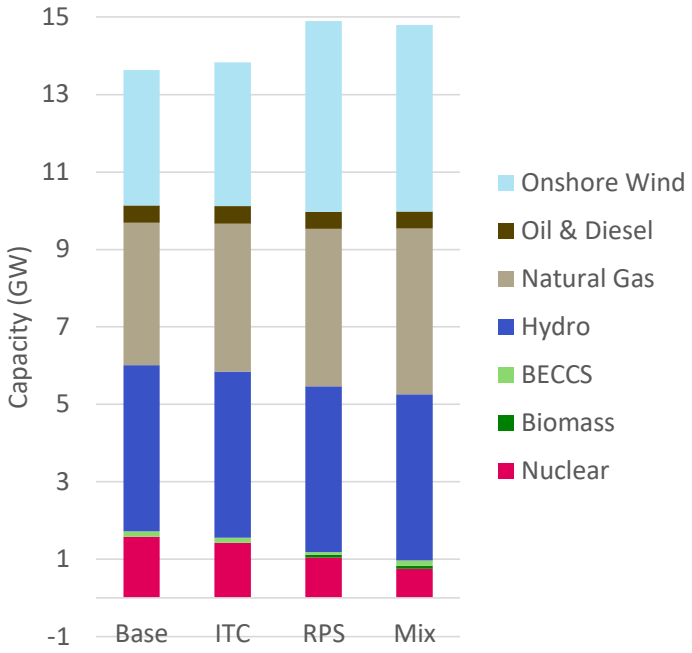
- Higher Capacity
- More Onshore Wind
- More Land Use





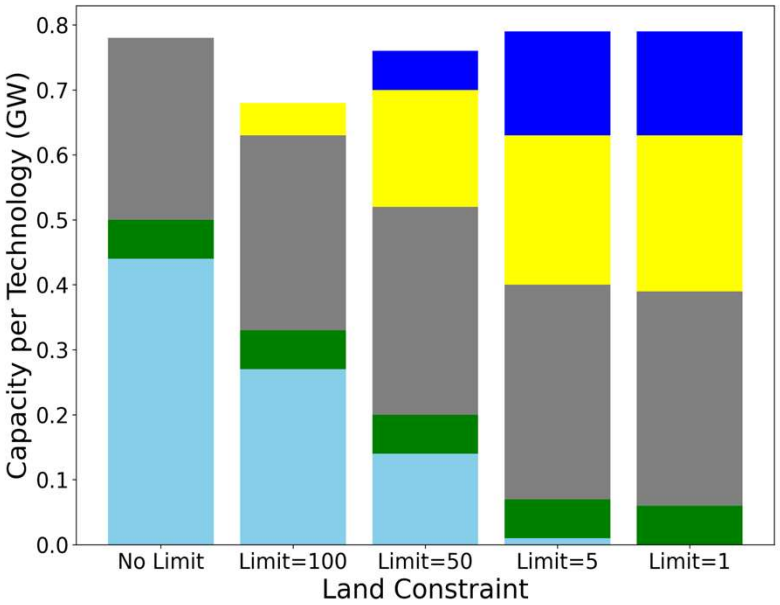
How different policies impact PEI's future Energy and Land Use- ITC & RPS

- Always Higher Capacity
- More Onshore Wind

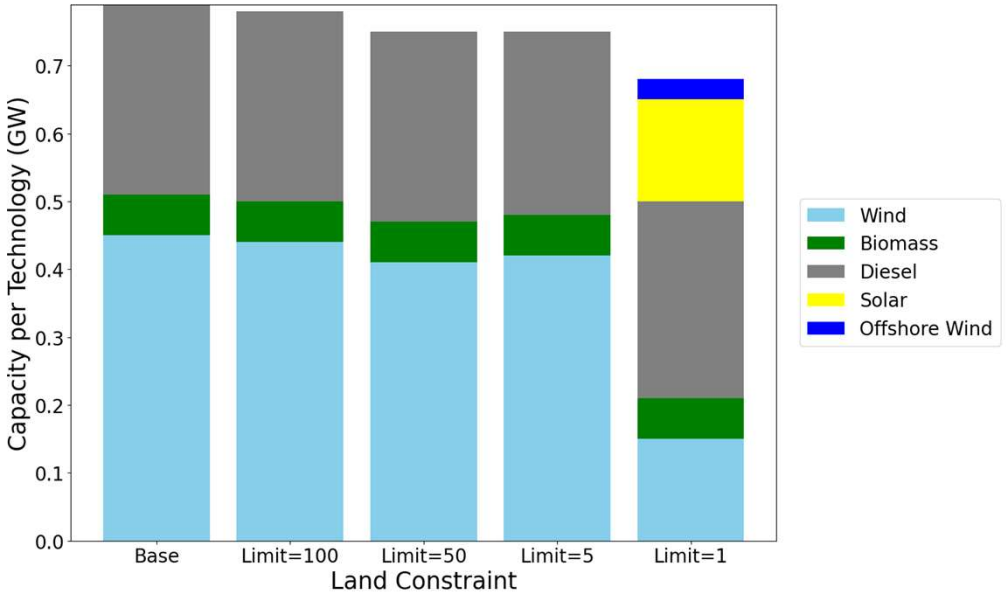


Onshore Wind and Cropland Colocation

- Offshore wind in the conventional scenario
- Slightly more diesel



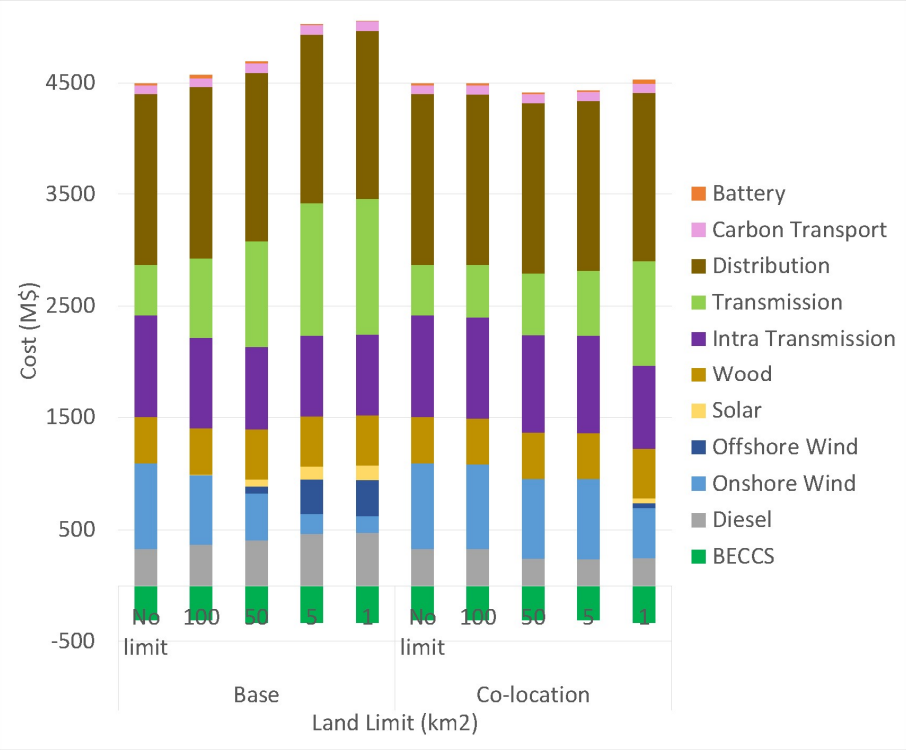
Conventional Scenario



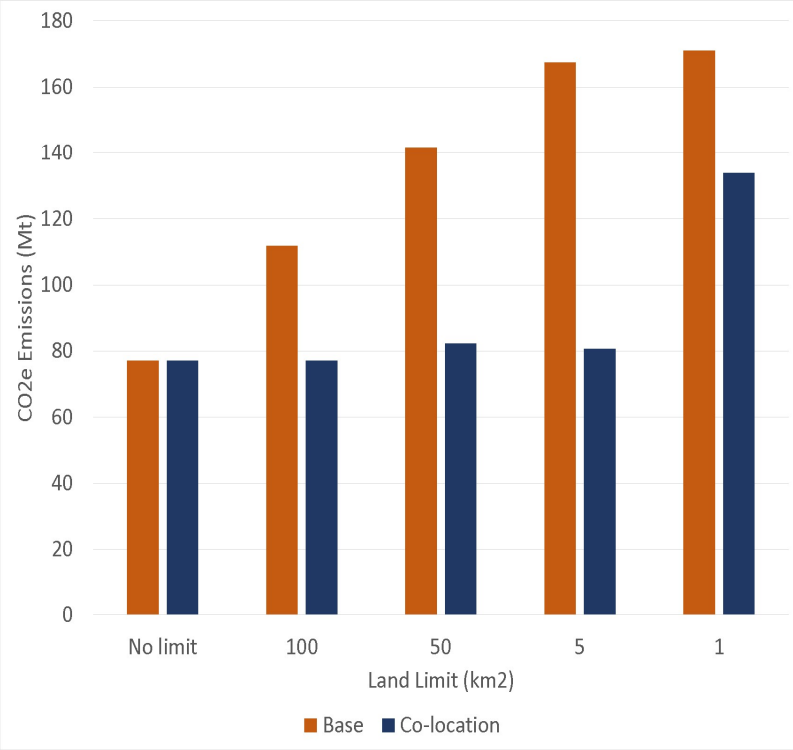
Colocation Scenario

Onshore Wind and Cropland Colocation

- Lower Costs



- Lower Emissions





Conclusions:

Constraining Land Covers

1. Within the land categories examined in this study, **cropland** creates the **largest modeled constraint** on future **onshore wind deployment**.
2. PEI is the **most affected** by this impact

Spacial Enhancement Benefits

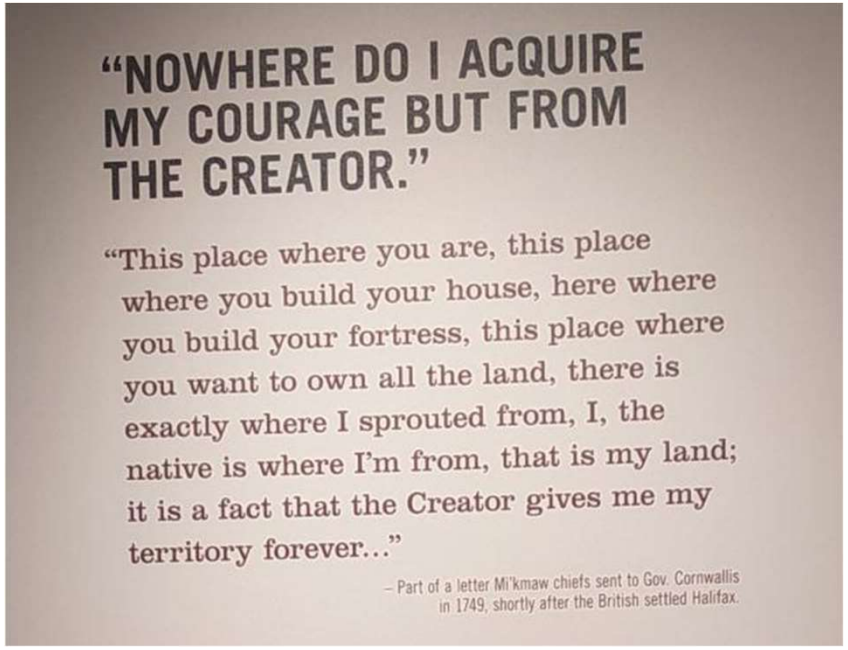
1. Integrating grid connection costs helps **minimize overall system development costs** (including grid connection costs)
2. Improving the technology resolution increases the overall **system's capacity factor**, and **reduces costs**

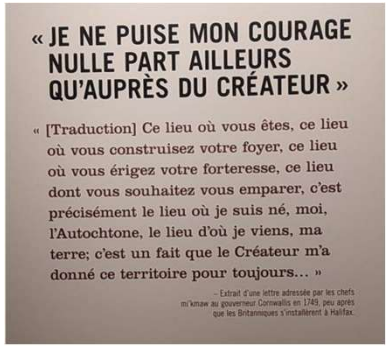
Policy Implications

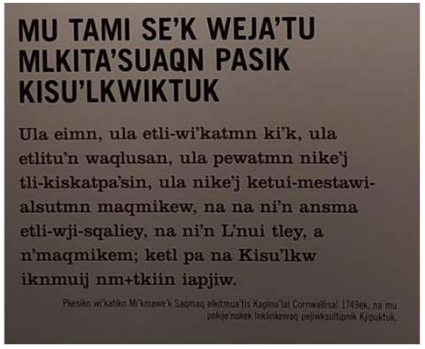
1. Business-as-usual as well as **policy scenarios** (ITC and RPS) result in **higher land use**

Colocation Modeling

1. Modeling onshore wind and agriculture colocation **reduces** the reliance on **expensive** and **emissions-intensive** generation **technologies**
2. Colocation modeling provides stronger support to decision-makers







THANK YOU

- Army Museum Halifax Citadel, May 2023

CONTRIBUTIONS SUMMARY

Conceptual Advancement

- The work demonstrates how land-intensity definitions and spatial assumptions can alter the interpretation of wind development in energy-system planning

Methodological Contributions

- The developed tool allows for studying space-related scenarios such as colocation by integrating spatially defined technologies and land uses into energy system optimization models.

Applied Case Studies

- The case-study provides evidence on how land availability, wind resource quality, clean-energy policies, and cropland assumptions influence technology choices, electricity generation, imports, emissions, costs, and land-use outcomes.

Policy and Planning Applications

- The work's insights on current modeling and policy can support more transparent planning discussions around renewable energy siting, agricultural land protection, and long-term decarbonization strategies by developing spatially enhanced energy models.

Validation- Choosing the Model

- Model feature comparison(with 4 other models), based on:

Scenario-based cost optimization

Existing infrastructure

Scenario management

Open-source license

Multiple regions

Multiple sectors

Flexible temporal resolution

Multiple investment periods

Documented codebase

Commercial and open-source solvers

User documentation, Active development, Python-based platform, Established use

Validation- Model's Scientific Rigor

- TEMOA publications
- ACES national and governmental reports and comparisons
- Multi-Model Comparison Forum's national initiative by EMH
- Expansion for Canada (CANOE initiative)

Validation- Input Data

- Wind Speed >> Global Wind Atlas
- Land Covers >> Cross-referencing with existing wind farms
- Land Intensities >>

Comparison across literature

Spatial comparison with OpenStreetMap

Cross-reference with government-published and other content

- Always using widely-known, publicly available and government-issued data sources

Validation- Model Outputs

- Edge cases

Table 11. Different land intensity values used in edge case validation		
Land intensity (km^2/GW)	Basis	Role in validation
1	Spacing-exclusive footprint	Lowest footprint-only/colocation edge case.
6.5	Spacing-exclusive footprint	Literature-based footprint-only value for wind without full spacing.
15.83	Project-level spacing-inclusive estimate	Lower spacing-inclusive estimate from the Summerside validation case.
368.3	Literature spacing-inclusive estimate	Primary spacing-inclusive onshore wind land intensity used in the base land accounting.
20,000	Rounded permitted-area estimate	Upper edge case based on the CODERS-derived PEI estimate.